Lecture 1

Clinical Engineering:

Defining the Profession and Its Role in Modern Healthcare

Shaping safer, smarter healthcare through engineering and technology management

Terminology Challenges

- Lack of consistent terminology still creates confusion
- **Issues span:** patient care, equipment integration, human factors, cybersecurity, financial improvement
- Always opportunities to improve health system performance

Healthcare Technology Management (HTM)

- Diverse professionals support clinicians with devices, software, and systems
- Broadly known as Healthcare Technology Management (HTM)Discipline combines:
  - Patient safety, Medical technology, Financial stewardship
- Application of engineering principles (analysis, systems principles) to improve healthcare

Healthcare Technology Management (HTM)

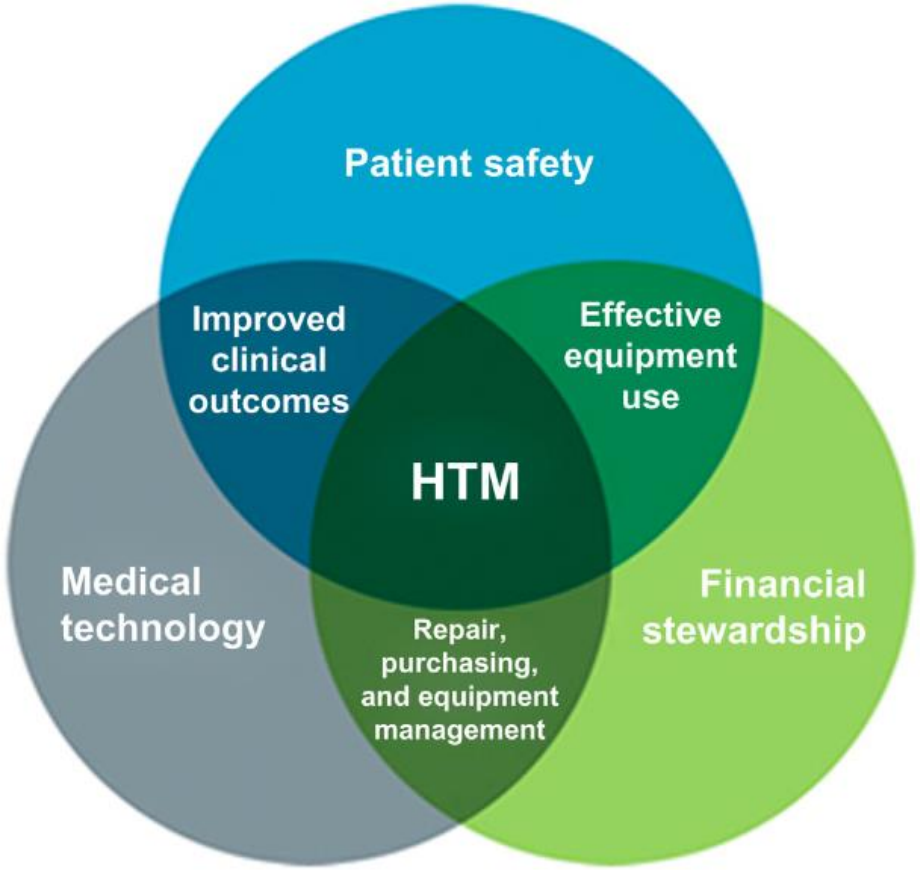


Figure 1.1 The healthcare technology management profession. *AAMI, used with permission.*

Definition of Clinical Engineering (ACCE)

- *"A Clinical Engineer is a professional who supports and advances patient care by applying engineering and managerial skills to health care technology."*



Clinical Engineer vs Biomedical Engineer

- Term clinical engineer **NOT** universally used
- Often interchanged with biomedical engineer (BME)

- Academic programs in U.S. → degrees in BME (not clinical engineering)
  Hospitals:
  - Departments named Clinical Engineering may employ staff with BME titles
  - Departments named BME may employ staff with Clinical Engineer titles

Biomedical Engineering Focus Areas (BMES)

Neural engineering	Bionanotechnology	Systems physiology	Rehabilitation engineering	Orthopedic bioengineering
Medical imaging	Clinical engineering	Cellular, tissue, and genetic engineering	Bionics	Biomechanics
	Bioinstrumentation	Biomechatronics	Biomedical electronics	

Biomedical Engineering vs Clinical Engineering

- BME = broad discipline
  - (artificial limbs → cellular research)

- Clinical engineering = more narrowly defined focal area within BME

WHO Terminology

- WHO (2017): “Biomedical engineering” includes similar disciplines, e.g.
  - Medical engineering
  - Electromedicine
  - Bioengineering
  - Medical and biological engineering
  - Clinical engineering

What is Clinical Engineering?

- A specialty within Biomedical Engineering.
- Focuses on the application and management of medical technology in healthcare settings.
- Clinical engineers:
  1. Work mainly in hospitals and health systems.
  2. Support the entire life cycle of medical devices (acquisition → maintenance → safety → retirement).
  3. Act as a bridge between technology, clinicians, and patients to ensure safe and effective care.

So, after all these inconsistent or semi-consistent terms, what is CCE?

- In Gaza (and many developing regions), clinical engineering is often the first thing people think of when hearing biomedical engineering or medical device engineering.

- The role usually refers to:
  - Maintenance and calibration of medical devices.
  - Management of hospital equipment and technology.
  - Consultancy and administrative roles related to medical equipment planning and safety.

HTM career Ladder

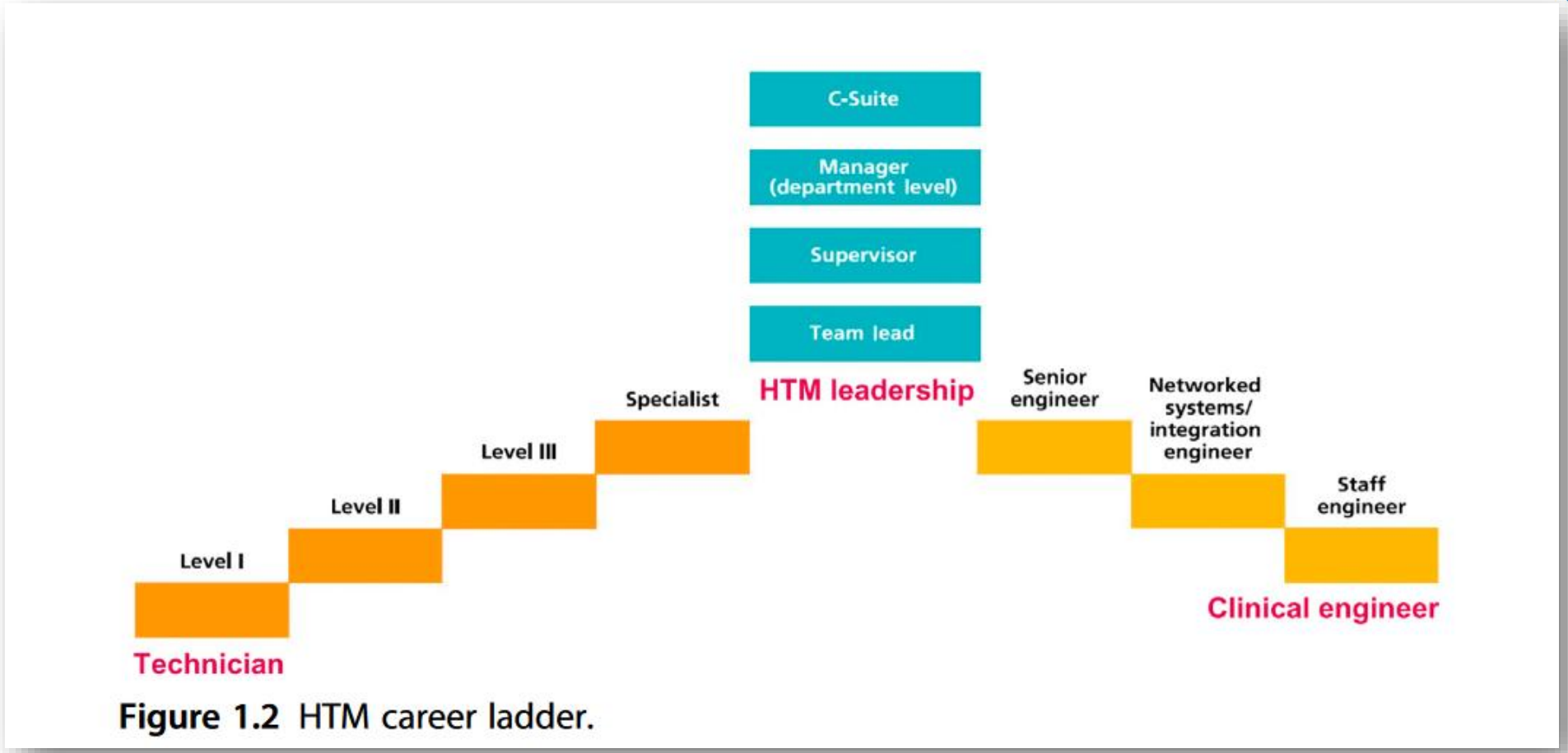
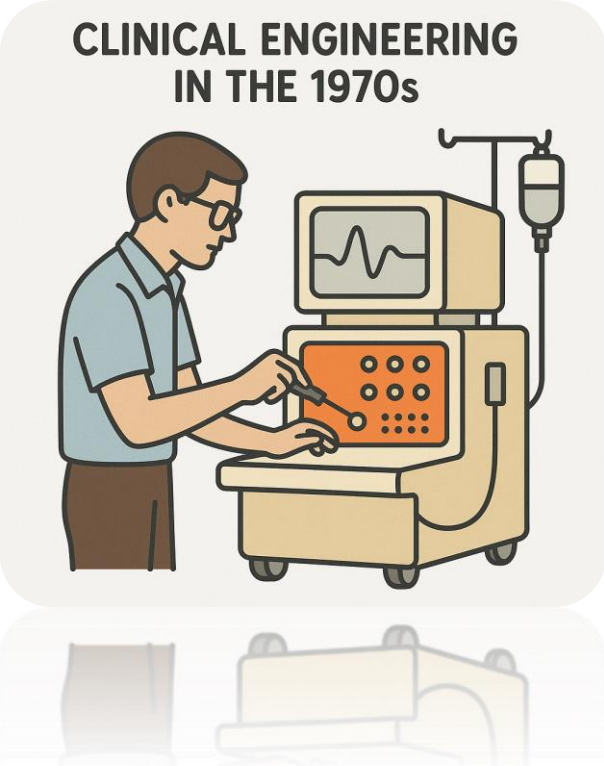


Figure 1.2 HTM career ladder.

Historical perspective



Origins and Early Awareness
(Late 1960s–1970s)

Expansion and Regulation
(1980s–1990s)

Professionalization and Recognition
(2000s)

IT, Data, AI, and New Specialties
(2009–now)

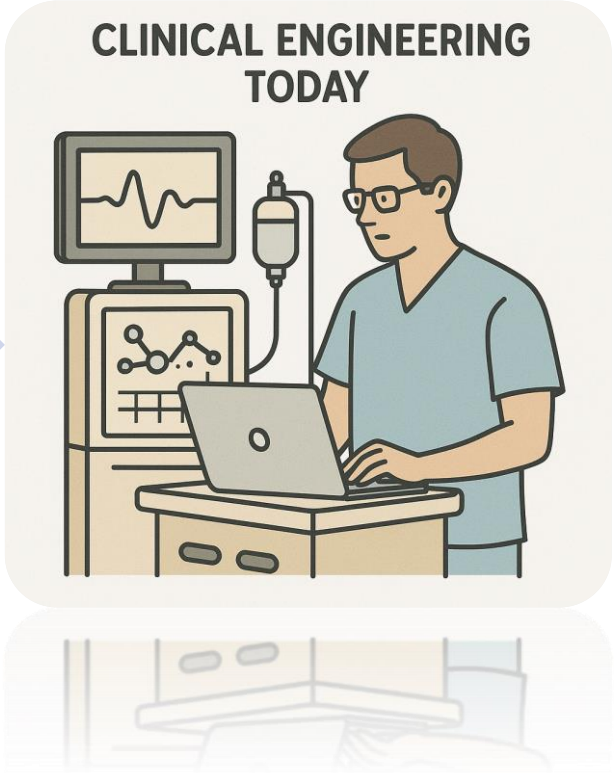
Historical perspective

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Historical perspective : Origins and Early Awareness (Late 1960s–1970s)

- **1960s** – Clinical engineering emerged as a biomedical engineering (BME) specialty focused on hospital technology implementation beyond repair/maintenance.
- **1971** – Ralph Nader published an article in *Ladies' Home Journal* claiming high rates of hospital electrocutions → triggered national concern for hospital safety.
- **Early 1970s** – Growth of community hospitals and increased use of monitoring and imaging technologies.
- **1970** – Certification for Biomedical Equipment Technicians (BMET) was introduced.
- **1973** – Association for the Advancement of Medical Instrumentation (AAMI) created a certification program for clinical engineers.

Expansion and Regulation (1980s–1990s)

- **1980s** – Hospital equipment inventories grew dramatically; pressure to reduce healthcare costs → expansion of hospital technical staff.
- **1984** – NFPA 99 (Health Care Facilities Code) established safety regulations addressing fire, explosion, and electrical hazards → clinical engineers became involved in compliance and safety.
- **1990** – Safe Medical Devices Act mandated hospitals to identify and report device problems. Hiring of biomedical engineers (with engineering degrees) expanded, strongly driven by the Veterans Administration (VA) hospital system.

Professionalization and Recognition (2000s)

- **Early 2000s** – Leaders such as Joseph Bronzino, Joseph Dyro, Malcolm Ridgway, and Yadin David promoted clinical engineering as a distinct engineering discipline, emphasizing formal engineering education (BSc/MSc).
- **2005** – National Academy of Engineering + Institute of Medicine report recommended applying systems engineering, human factors, supply chain, and IT principles to healthcare delivery.
- **2000s onward** – Expansion of academic programs, publications, advocacy, and differentiation between hospital technicians and engineers.

IT, Data, and New Specialties (2009–2010s)

- **2009** – American Recovery and Reinvestment Act (ARRA) with the HITECH Act incentivized electronic medical record (EMR) adoption.
- **2012** – Deadline for electronic health record implementation → clinical engineers took roles in managing data, integrating medical devices with IT systems, and ensuring cybersecurity.
- **2010s** – Emergence of network clinical engineering and cybersecurity specialties. Clinical engineers began working more closely with hospital IT staff.

Certification and credentialing

- **CCE (Certified Clinical Engineer):**
  - ACCE/US-Canada board; written + oral exam; renewal every 3 yrs.
- **CHTM (Certified Healthcare Technology Manager) :**
  - AAMI; 100-question exam on management; renewal every 3 yrs.
- **CBET (Certified Biomedical Equipment Technician) :**
  - AAMI; 165-question exam on devices/IT; renewal every 3 yrs.

Roles of the Clinical Engineer (1)

1. Identify and assess institutional technology needs
2. Plan, evaluate, and manage equipment purchases
3. Oversee installation and integration of medical devices
4. Manage device lifecycle: use, maintenance, end-of-life
5. Ensure safe integration with networks & electronic medical records

Roles of the Clinical Engineer (2)

6. Apply systems thinking to improve hospital processes

7. Support clinicians with training & translation of technical concepts

8. Ensure regulatory compliance and quality assurance

9. Manage human resources related to medical technology

10. Provide strategic vision for future technology adoption

11. Oversee device cybersecurity and data protection

Identify technology needs

Scenarios

- **A hospital is opening a new ICU →** the clinical engineer evaluates what ventilators and monitors are required.

- **Emergency department expands →** engineer identifies need for extra defibrillators and portable ultrasound.

Plan & manage equipment purchases

Scenarios

- **Before buying** an MRI, the engineer compares vendors, pricing, and performance to recommend the best option.

- **Hospital considers robotic surgery system** → engineer prepares cost–benefit analysis for leadership.

Oversee installation & integration

Scenarios

- **New infusion pumps arrive →** the engineer ensures correct installation and connection to hospital networks.

- **CT scanner delivered →** engineer coordinates room shielding checks and IT integration.

Manage device lifecycle

Scenarios

- **A 10-year-old ultrasound machine is breaking down →** the engineer decides whether to repair, upgrade, or dispose.

- **Aging anesthesia machines →** engineer phases them out and plans upgrade schedule.

Integrate with EMR & networks

Scenarios

- **A new patient monitor must send data directly into the electronic medical record →** the engineer configures the interface.

- **Mobile X-ray →** engineer ensures images automatically transfer to PACS system.

Apply systems thinking

Scenarios

- **Operating rooms have delays due to equipment sharing →** the engineer redesigns scheduling to optimize device availability.

- **Multiple departments using same sterilizers →** engineer introduces tracking system to reduce bottlenecks.

Educate clinicians on technology

Scenarios

- **Nurses receive new ventilators** → the engineer trains them on safe operation and troubleshooting.

- **Doctors adopt handheld ultrasound** → engineer runs demo sessions and troubleshooting workshops.

Ensure regulatory compliance

Scenarios

- **A device inspection finds missing safety checks** → the engineer updates procedures to meet standards.

- **FDA*/MDR* audit scheduled** → engineer prepares documentation for all high-risk devices.

Quality assurance in technology management

Scenarios

- The engineer implements **routine calibration/testing** to reduce medical device errors.

- Introduces **checklist** system to track device performance and reduce failure rates.

Human capital management

Scenarios

- **A hospital hires new biomedical technicians** → the engineer organizes their training and supervision.

- **Staff shortage → engineer** reallocates team and provides cross-training to cover gaps.

Strategic vision for technology adoption

Scenarios

- The engineer **advises** hospital leadership on adopting AI-based diagnostic tools for radiology.

- **Recommends** hospital-wide telemedicine integration to improve rural patient access.

Cybersecurity & data protection

Scenarios

- **A ransomware attack targets infusion pumps →** the engineer works with IT to secure devices and protect patient data.

- **Wireless pacemakers in use →** engineer ensures encryption and secure monitoring.

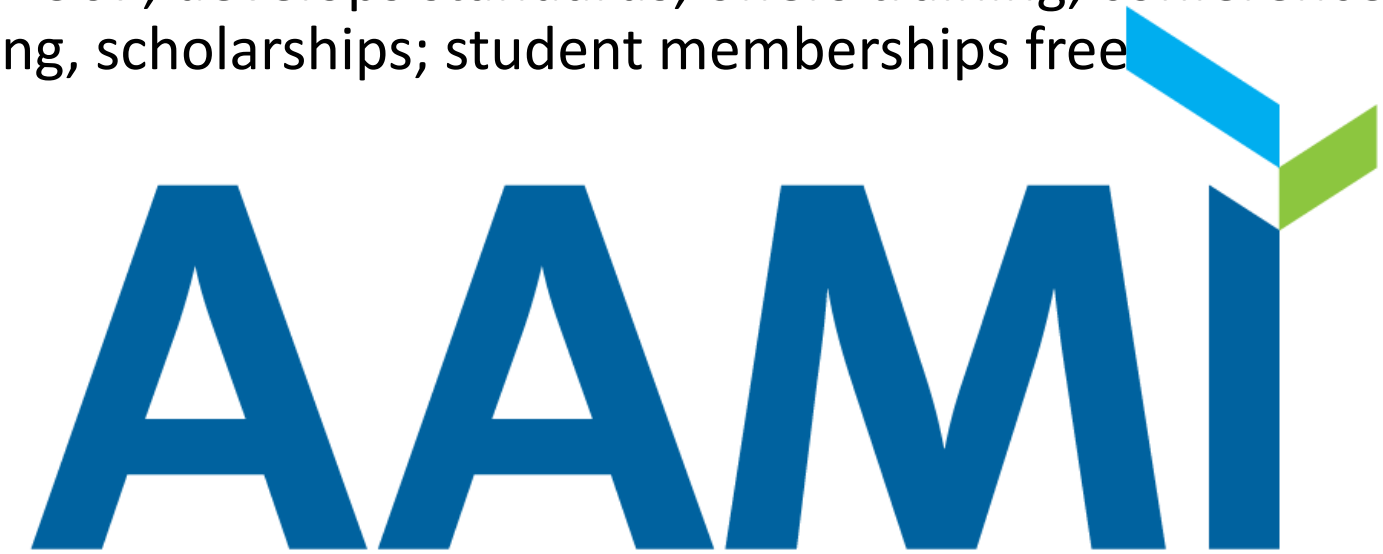
Societies and collaboration

- **American College of Clinical Engineering (ACCE)**
  - Founded 1990; sets competence standards, promotes excellence, supports education, advocacy, awards, scholarships, webinars.



Societies and collaboration

- **Association for the Advancement of Medical Instrumentation (AAMI)**
  - Founded 1967; develops standards, offers training, conferences, publications, networking, scholarships; student memberships free



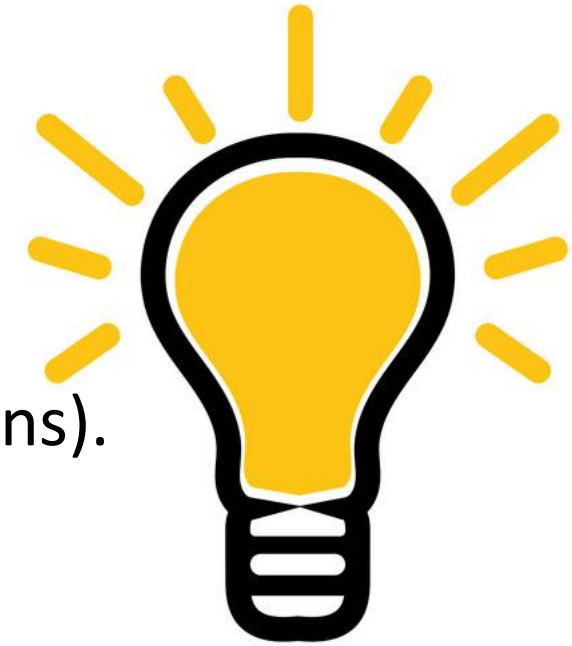
Societies and collaboration

- **American Society for Health Care Engineering (ASHE)**
  - Part of AHA; focuses on hospital facility design, maintenance, operations; offers education, advocacy, and networking.



Future of Clinical Engineering

- Hard to predict technology, but change is constant.
- Clinical engineers must stay proactive and adaptable.
- Continuous learning is essential (new tech, new regulations).
- Growing role in home-based devices & digital health.
- Success depends on collaboration with all stakeholders.



Think with me !? 🤔

- Do you think there are **real technical differences** between the roles of biomedical engineers and clinical engineers in our environments?

- Do you think **specialization** in the field is better, or is it more valuable to remain a **generalist** (e.g., biomedical/clinical combined) in order to be multi-purpose and fill different gaps?

- Reply to me with your thoughts !


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Thank you